

# SFT V11.2 Results Ledger

Retarded Dent Field Program - Computational Ledger A to N2  
Prepared for Shaun Fosmark. Generated 2026-05-10.

Primary calibration: Born equilibrium preservation passed; Born relaxation mechanism seeded; Born-rule derivation remains open.

## Purpose and use

This ledger freezes the V11.2 computational run so it can be reuploaded later without losing the hierarchy of results. It is not a paper draft and not a claim-expansion document. It is a guardrail against claim drift: each item records what passed, what was seeded, what failed or was superseded, and what remains open.

## Claim tier summary

| Tier                                  | Contents                                                                                                                                                                                                                                                                | Current status                                             |
|---------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------|
| Closed / strong representation passes | Interference, shot-building, trajectory guidance, decoherence, uncertainty scale, Born equivariance, de Broglie carrier, pure-state quantum potential, Bell/no-signaling grammar, spin-holonomy Bell bridge, boundary channel consistency, action-phase current bridge. | Strong V11.2 representation-level architecture.            |
| Seeded but not derived                | Mixing-linked Born relaxation mechanism; native current via maintained action phase / clock holonomy; boundary measurement channel grammar.                                                                                                                             | Mechanism seeds requiring deeper theory and/or robustness. |
| Still open                            | Born-rule derivation; V11.1B settlement equation -> action phase S; detector microphysics; full relativistic boundary update dynamics; physical creation/update of boundary singlets.                                                                                   | Explicitly open. Do not promote to closed claims.          |

# Master ledger: V11.2A through V11.2N2

| ID | Question / sector                             | Outcome                       | Key result                                                                                                                                                          | Claim level                                                      | Guardrail / open item                                                        |
|----|-----------------------------------------------|-------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------|------------------------------------------------------------------------------|
| A  | Quantitative double-slit dent kernel          | PASS                          | Two-path correlated dent phase kernel reproduces modulation, fringe spacing, scaling, and $\beta=2\pi/\lambda$ ; aperture/local guardrails fail.                    | QUANTITATIVE_KERNEL_PASS_NOT_PARENT_DERIVATION                   | Kernel grammar pass; V11.1B parent derivation remains open.                  |
| B  | Settlement parent support for kernel          | PASS + CHECK                  | Clocked settlement Green residual carrying de Broglie/action phase reproduces target modulation (corr 0.999645). Short-memory/decoherence behavior remains a check. | PARENT_KERNEL_SUPPORT_PASS_CARRIER_NOT_DERIVED                   | Settlement can support phase kernel; carrier and Born rule not derived here. |
| C  | Localized shots build density                 | PASS                          | 10,000 localized screen events generated; ensemble histogram corr 0.998351 to sampled V11.2 density.                                                                | SHOT_BUILDING_VISUALIZATION_NOT_BORN_DERIVATION                  | Sampling from density is not a Born derivation.                              |
| D  | Trajectory dent guidance                      | PASS                          | Deterministic settlement-guided trajectories populate coherent field density (corr 0.994644); free/local guardrails worse.                                          | TRAJECTORY_GUIDANCE_PASS_NOT_BORN_OR_CARRIER_DERIVATION          | Does not derive Born rule or carrier.                                        |
| E  | Which-path decoherence trajectory gate        | PASS                          | Detector overlap continuously suppresses cross-term; visibility/complementarity errors $\sim 1e-5$ ; trajectory histograms follow partial coherence densities.      | TRAJECTORY_DECOHERENCE_PASS_NOT_BORN_OR_MEASUREMENT_DERIVATION   | Detector microphysics and Born rule remain open.                             |
| F  | Single-slit uncertainty scale                 | PASS                          | Spectrum satisfies $\Delta x \Delta p \geq \hbar/2$ (1.161483); inverse-width scaling slope -1.0; guided trajectories follow density.                               | UNCERTAINTY_SCALE_PASS_NOT_BORN_OR_CARRIER_DERIVATION            | Uncertainty-scale behavior only; $\lambda=h/p$ and Born rule remain open.    |
| G  | Born equivariance pressure test               | PASS + GUARDRAIL              | Born-initialized ensembles remain Born-like (min corr 0.992065); non-Born does not instantly become Born.                                                           | EQUIVARIANCE_PASS_BORN_ORIGIN_NOT_DERIVED                        | Born equilibrium preservation passed; origin/relaxation open.                |
| H  | de Broglie carrier/action phase origin        | PASS                          | Lorentz-projected Compton/action phase selects $k=p/\hbar$ ; worldline phase and dispersion/group velocity gates pass to roundoff.                                  | CARRIER_ORIGIN_ACTION_PHASE_PASS_NOT_FULL_DYNAMICAL_DERIVATION   | Kinematic/action-phase selection; deeper momentum dynamics and Born open.    |
| I  | Quantum potential / dent-force reconstruction | PASS (pure states)            | Pure single/double slit material acceleration matches $-(1/k) \nabla Q[\rho]$ ; decohered global-Q case diagnostic only.                                            | QUANTUM_POTENTIAL_FORCE_RECONSTRUCTION_PASS_PURE_STATE_ONLY      | Mixed/density-matrix measurement dynamics not solved.                        |
| J  | Bell/no-signaling boundary grammar            | PASS + GUARDRAIL              | Boundary singlet kernel gives $CHSH=2\sqrt{2}$ , no-signaling, Werner threshold; local dent guardrail stays at $CHSH \leq 2$ .                                      | BELL_BOUNDARY_GRAMMAR_PASS_NOT_DERIVED_DYNAMICALLY               | Boundary kernel represented, not dynamically derived.                        |
| K  | Singlet spin-holonomy bridge                  | PASS                          | V11.9 singlet spinor/projectors generate V11.2J Bell kernel with rotational invariance and no-signaling.                                                            | SINGLET_HOLONOMY_BRIDGE_PASS_NOT_BOUNDARY_DYNAMICS_DERIVED       | Physical singlet creation/update dynamics remain open.                       |
| L  | Boundary measurement/no-signaling channels    | PASS                          | Zero-total spin projector selects singlet; measurement order invariance and local detector no-signaling pass to roundoff.                                           | BOUNDARY_MEASUREMENT_CHANNELS_PASS_DYNAMICALLY_NOT_DERIVED       | Representation/channel consistency only; physical dynamics open.             |
| M  | Naive Born relaxation attempt                 | FAILED CONTROL / INCONCLUSIVE | Stationary guardrail passed, but multimode Born-control failed (corr 0.180934), invalidating relaxation interpretation.                                             | BORN_RELAXATION_MECHANISM_SEED_NOT_DERIVATION (superseded)       | Superseded by M2/M3; lesson: no relaxation claim without Born preflight.     |
| M2 | Equivariance-safe Born relaxation retest      | PASS PREFLIGHT + DIAGNOSTIC   | Adaptive unclipped torus flow fixed Born preflight (corr 0.980289); non-Born H decrease mild and diagnostic.                                                        | BORN_PREFLIGHT_PASS_RELAXATION_DIAGNOSTIC_INTERPRETABLE          | Relaxation not closed; only interpretable after preflight.                   |
| M3 | Mixing-linked Born relaxation scan            | PASS_MECHANISM                | Preflight pass fraction 0.833333; $H_f/H_0$ decreases monotonically with mixing proxy (Spearman -1.0); best ratio 0.390791.                                         | MIXING_LINKED_BORN_RELAXATION_MECHANISM_SEED                     | Mechanism seed, not Born-rule derivation.                                    |
| N  | Native lag-current bridge cascade             | USEFUL FAILURE                | Density-only lag guardrail passed (corr 0.0); action-phase bridge initially failed due to phase-extraction/scoring issues; cascade stopped.                         | NATIVE_LAG_CURRENT_BRIDGE_NOT_CLOSED (superseded by N2)          | Shows density dents alone are insufficient; superseded by N2 audit.          |
| N2 | Action-phase / quantum-current bridge audit   | PASS                          | $v_Q = \text{Im}(\psi^* \nabla \psi) /  \psi ^2$ matches grad S in controlled and node-masked domains; errors $2.6e-13$ to $1.6e-5$ .                               | ACTION_PHASE_CURRENT_BRIDGE_VALIDATED_CONTROLLED_AND_NODE_MASKED | Still does not derive S from V11.1B settlement dynamics.                     |

# Known failures, supersessions, and lessons

| Item                 | What happened                                                                                                    | Resolution / lesson                                                                                                                                               |
|----------------------|------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| V11.2M               | Naive Born relaxation run failed the Born-control: Born-initialized multimode ensemble did not remain Born-like. | Do not interpret relaxation if Born preflight fails. M2/M3 supersede M for Born relaxation status.                                                                |
| V11.2N               | Density-only lag guardrail passed, but action-phase bridge initially failed in gnarly multimode case.            | Failure was traced to phase extraction / branch cuts / constant-field scoring. N2 v2 validates action-phase current bridge on controlled and node-masked domains. |
| Short-memory B check | Short memory/decoherence suppression was marked CHECK even though parent kernel support passed.                  | Keep as a caution: B supports correlated phase grammar but does not derive carrier, Born, or complete decoherence dynamics.                                       |

# Dependency map

| Dependency                          | Feeds into                           | Current relationship                                                                                  |
|-------------------------------------|--------------------------------------|-------------------------------------------------------------------------------------------------------|
| V11.1B lag-settlement residual      | V11.2B, V11.2N/N2, future V12 bridge | Provides parent residual language; does not yet derive maintained action phase S.                     |
| V11.9 spin holonomy                 | V11.2K/L Bell boundary sector        | Supplies SU(2) singlet/projector machinery that generates the V11.2J Bell kernel.                     |
| Maintained Compton/action phase     | V11.2H/N2                            | Selects de Broglie carrier and reconstructs quantum current as grad S; settlement-to-S dynamics open. |
| Boundary ledger / holographic layer | V11.2J/K/L                           | Required for Bell correlations and no-signaling; physical update dynamics open.                       |

## Detailed verdict tables

These tables preserve the actual gate/status/value rows from the verdict CSVs used in this session. The claim\_level row is the controlling claim label for each script.

### V11.2A - Quantitative double-slit dent kernel

| Gate                                        | Status                                         | Meaning                                                                                                       | Value     |
|---------------------------------------------|------------------------------------------------|---------------------------------------------------------------------------------------------------------------|-----------|
| correlated_kernel_matches_target_modulation | PASS                                           | two-path correlated dent phase kernel reproduces target modulation                                            | 1         |
| correlated_kernel_fringe_spacing            | PASS                                           | main config fringe spacing matches $\Delta y = \lambda L / d$                                                 | 0.019     |
| scaling_sweep_fringe_spacing                | PASS                                           | correlated kernel preserves $\Delta y$ scaling over $d$ , $L$ , $\lambda$ , and $a$ sweeps                    | 0.0373214 |
| aperture_and_local_guardrails               | PASS_GUARDRAIL                                 | aperture-only and local-slit dents do not pass the fringe-train spacing gate                                  | 0         |
| beta_fit_recovers_debroglie_phase           | PASS                                           | best-fit beta recovers $\beta = 2\pi/\lambda$                                                                 | 1         |
| claim_level                                 | QUANTITATIVE_KERNEL_PASS_NOT_PARENT_DERIVATION | correlated dent phase grammar can reproduce double-slit law; V11.1B settlement parent derivation remains open |           |

### V11.2B - Settlement parent support for kernel

| Gate                                    | Status                                         | Meaning                                                                                                       | Value    |
|-----------------------------------------|------------------------------------------------|---------------------------------------------------------------------------------------------------------------|----------|
| clocked_settlement_green_matches_target | PASS                                           | coherent settlement residual carrying de Broglie/action phase reproduces target modulation and spacing        | 0.999645 |
| phase_slope_recovers_debroglie_beta     | PASS                                           | parent phase carrier has $\beta = 2\pi/\lambda$ with correct path-difference slope                            | 1        |
| dc_and_incoherent_guardrails            | PASS_GUARDRAIL                                 | DC settlement residual and incoherent power sum do not generate the fringe train                              | 0        |
| short_memory_suppresses_visibility      | CHECK                                          | short residual memory/decoherence suppresses the coherent fringe visibility                                   | 0.999965 |
| coherence_memory_threshold              | PASS                                           | scan identifies memory time needed for high-correlation two-path residual                                     | 1.61268  |
| claim_level                             | PARENT_KERNEL_SUPPORT_PASS_CARRIER_NOT_DERIVED | settlement residual can support the correlated phase kernel, but de Broglie carrier and Born rule remain open |          |

### V11.2C - Localized shots build density

| Gate                              | Status                                          | Meaning                                                                                                    | Value     |
|-----------------------------------|-------------------------------------------------|------------------------------------------------------------------------------------------------------------|-----------|
| localized_hits_generated          | PASS                                            | script generated localized screen events                                                                   | 1.000e+04 |
| ensemble_converges_toward_density | PASS                                            | final histogram correlates with the sampled V11.2 density; low nshots are noisy by design                  | 0.998351  |
| claim_level                       | SHOT_BUILDING_VISUALIZATION_NOT_BORN_DERIVATION | localized hits build the predicted fringe density, but sampling from density is not a Born-rule derivation |           |

### V11.2D - Trajectory dent guidance

| Gate                                        | Status                                                  | Meaning                                                                                                                            | Value    |
|---------------------------------------------|---------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------|----------|
| trajectory_guidance_populates_field_density | PASS                                                    | deterministic settlement-guided trajectories populate the propagated coherent field density                                        | 0.994644 |
| free_and_local_guardrails_worse             | PASS_GUARDRAIL                                          | free rays and local-only edge forces do not match the coherent density as well as settlement guidance                              | 0.405795 |
| field_density_near_fraunhofer_target        | PASS_SCALE                                              | guided detections are also broadly correlated with the far-field double-slit target                                                | 0.77154  |
| claim_level                                 | TRAJECTORY_GUIDANCE_PASS_NOT_BORN_OR_CARRIER_DERIVATION | localized trajectories guided by coherent dent phase-current build the field density; de Broglie carrier and Born rule remain open |          |

### V11.2E - Which-path decoherence trajectory gate

| Gate                                           | Status                                                         | Meaning                                                                                                          | Value     |
|------------------------------------------------|----------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------|-----------|
| coherent_gamma1_trajectory_density             | PASS                                                           | gamma=1 guided trajectories populate the coherent two-path density                                               | 0.99078   |
| which_path_gamma0_suppresses_cross_term        | PASS                                                           | gamma=0 removes the coherent cross-term / fringe visibility in the field density                                 | 0         |
| visibility_tracks_detector_overlap             | PASS                                                           | field visibility cross-fraction tracks detector overlap gamma                                                    | 1.326e-05 |
| complementarity_relation                       | PASS                                                           | $V^2 + D^2$ remains near one using V from dent cross-term and $D = \sqrt{1 - \gamma^2}$                          | 1.492e-05 |
| trajectory_histograms_follow_partial_coherence | PASS                                                           | guided trajectory histograms follow their corresponding partially coherent densities across gamma sweep          | 0.99078   |
| claim_level                                    | TRAJECTORY_DECOHERENCE_PASS_NOT_BORN_OR_MEASUREMENT_DERIVATION | which-path overlap suppresses dent-guided interference continuously; Born rule and detector dynamics remain open |           |

## V11.2F - Single-slit uncertainty scale

| Gate                                            | Status                                                | Meaning                                                                                                              | Value    |
|-------------------------------------------------|-------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------|----------|
| uncertainty_bound_spectrum                      | PASS                                                  | single-slit aperture spectrum satisfies $\Delta x \Delta p \geq \hbar/2$ within numerical tolerance                  | 1.16148  |
| inverse_width_scaling                           | PASS                                                  | momentum spread scales approximately as $\Delta p$ proportional to $1/\Delta x$                                      | -1       |
| trajectory_guidance_follows_single_slit_density | PASS                                                  | guided trajectory screen histograms follow propagated single-slit field densities across width sweep                 | 0.915086 |
| trajectory_momentum_scale                       | PASS_SCALE                                            | trajectory slopes carry a nonzero momentum-spread scale; exact spectral equality is not required in this toy         | 0.976578 |
| claim_level                                     | UNCERTAINTY_SCALE_PASS_NOT_BORN_OR_CARRIER_DERIVATION | single-slit dent/current architecture reproduces uncertainty-scale spread; Born rule and $\lambda = h/p$ remain open |          |

## V11.2G - Born equivariance pressure test

| Gate                                   | Status                                    | Meaning                                                                                                               | Value     |
|----------------------------------------|-------------------------------------------|-----------------------------------------------------------------------------------------------------------------------|-----------|
| born_equivariance_across_fields        | PASS                                      | Born-initialized guided ensembles remain distributed like their propagated dent/current densities                     | 0.992065  |
| born_H_function_stays_low              | PASS                                      | Born-initialized ensembles keep low coarse-grained H relative to $ \Psi ^2$                                           | 0.0111347 |
| double_slit_coherent_equivariance      | PASS                                      | coherent double-slit Born ensemble follows the propagated coherent two-path density                                   | 0.992279  |
| nonborn_does_not_instantly_become_born | PASS_GUARDRAIL                            | biased non-Born two-slit ensemble remains statistically distinguishable from Born at the screen                       | 0.277941  |
| coarse_grained_relaxation_diagnostic   | DIAGNOSTIC_OPEN                           | median $H_{\text{final}}/H_{\text{initial}}$ for non-Born cases; relaxation is not claimed closed in this toy         | 1.31759   |
| claim_level                            | EQUIVARIANCE_PASS_BORN_ORIGIN_NOT_DERIVED | dent guidance preserves Born equilibrium and respects non-Born guardrail; origin/relaxation of Born rule remains open |           |

## V11.2H - de Broglie carrier/action phase origin

| Gate                                                | Status                                                         | Meaning                                                                                                                      | Value     |
|-----------------------------------------------------|----------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------|-----------|
| lorentz_projected_compton_clock_selects_debroglie_k | PASS                                                           | Lorentz projection of maintained Compton/action phase gives $k = p/\hbar$ , the V11.2 kernel slope                           | 0         |
| worldline_phase_matches_proper_compton_clock        | PASS                                                           | $\omega t - k x$ along $x = vt$ equals $\omega_C \tau$                                                                       | 1.019e-15 |
| relativistic_dispersion_and_group_velocity          | PASS                                                           | $\omega^2 - c^2 k^2 = \omega_C^2$ and $d\omega/dk = v$                                                                       | 8.882e-16 |
| double_slit_phase_kernel_matches_v11_2a             | PASS                                                           | $\Delta \phi = k \Delta \ell$ uses $k = p/\hbar$ , yielding $\lambda = h/p$                                                  | 0         |
| alternative_carriers_fail_guardrails                | PASS_GUARDRAIL                                                 | raw Compton/lab Compton/no-spatial/nonrel carriers fail the required V11.2 phase slope somewhere in sweep                    | 0.56411   |
| claim_level                                         | CARRIER_ORIGIN_ACTION_PHASE_PASS_NOT_FULL_DYNAMICAL_DERIVATION | de Broglie carrier is selected by Lorentz-projected Compton/action phase; Born rule and deeper momentum dynamics remain open |           |

## V11.2I - Quantum potential / dent-force reconstruction

| Gate                                   | Status                                                      | Meaning                                                                                                                                 | Value    |
|----------------------------------------|-------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|----------|
| single_slit_pure_qforce_reconstruction | PASS                                                        | single-slit material acceleration matches $-(1/k) \text{ grad } Q[\rho]$                                                                | 0.968634 |
| double_slit_pure_qforce_reconstruction | PASS                                                        | coherent double-slit material acceleration matches quantum-potential force away from nodes                                              | 0.97414  |
| pure_state_linear_force_fit            | PASS                                                        | pure-state $a_{\text{material}}$ vs $a_Q$ fit has near-unit slope and high explanatory power                                            | 0.938253 |
| decohered_globalQ_guardrail            | DIAGNOSTIC                                                  | partially/incoherent cases are density-matrix/branch problems; simple global scalar $Q$ is not overclaimed                              | 0.715431 |
| claim_level                            | QUANTUM_POTENTIAL_FORCE_RECONSTRUCTION_PASS_PURE_STATE_ONLY | dent guidance is compatible with $Q[\rho]$ force geometry for pure states; Born origin, Bell, and full measurement dynamics remain open |          |

## V11.2J - Bell/no-signaling boundary grammar

| Gate                                            | Status                                          | Meaning                                                                                                                     | Value   |
|-------------------------------------------------|-------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------|---------|
| boundary_kernel_reproduces_singlet_correlations | PASS                                            | boundary singlet kernel gives $\text{CHSH}=2\sqrt{2}$ at canonical settings                                                 | 2.82843 |
| local_dent_guardrail_obeyes_bell_bound          | PASS_GUARDRAIL                                  | local shared-variable dent model does not violate CHSH                                                                      | 2       |
| no_signaling_marginals                          | PASS                                            | boundary correlations change joint outcomes but keep local marginals independent of distant setting                         | 0       |
| noise_threshold_matches_werner_limit            | PASS_SCALE                                      | Bell violation threshold occurs near $\eta=1/\sqrt{2}$                                                                      | 0.705   |
| claim_level                                     | BELL_BOUNDARY_GRAMMAR_PASS_NOT_DERIVED_DYNAMICS | SFT needs boundary-level correlation grammar for Bell while preserving no-signaling; local dent propagation is insufficient |         |

## V11.2K - Singlet spin-holonomy bridge

| Gate                                            | Status                                                     | Meaning                                                                                                                                 | Value     |
|-------------------------------------------------|------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|-----------|
| spinor_projectors_generate_boundary_kernel      | PASS                                                       | singlet state plus $\text{SU}(2)$ projectors gives $P(s,t)=1/4[1-\text{st}(a \cdot b)]$ over angle grid                                 | 5.551e-17 |
| random_3d_axis_rotational_invariance            | PASS                                                       | singlet holonomy result holds for random 3D measurement axes, not only planar angles                                                    | 6.661e-16 |
| chsh_tsirelson_from_singlet_holonomy            | PASS                                                       | singlet spinor/projector bridge gives $\text{CHSH}=2\sqrt{2}$                                                                           | 2.82843   |
| separable_and_local_guardrails_do_not_fake_bell | PASS_GUARDRAIL                                             | product, classical mixture, and local shared-variable guardrails stay at or below $\text{CHSH}=2$                                       | 2         |
| no_signaling_marginals_from_singlet_projectors  | PASS                                                       | local marginals remain 1/2 independent of distant measurement axis                                                                      | 1.110e-16 |
| claim_level                                     | SINGLET_HOLONOMY_BRIDGE_PASS_NOT_BOUNDARY_DYNAMICS_DERIVED | V11.9 spin-holonomy machinery generates the V11.2J boundary Bell kernel; physical creation/dynamics of the boundary singlet remain open |           |

## V11.2L - Boundary measurement/no-signaling channels

| Gate                                      | Status                                                  | Meaning                                                                                                                                          | Value     |
|-------------------------------------------|---------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|-----------|
| zero_total_spin_projector_selects_singlet | PASS                                                    | boundary zero-total spin-holonomy selection outputs singlet with $S_{\text{total}}^2=0$ when branch has nonzero overlap                          | 2.220e-16 |
| measurement_order_invariance              | PASS                                                    | Alice-first, Bob-first, and simultaneous spacelike projectors give same singlet joint probabilities                                              | 4.441e-16 |
| local_detector_channel_no_signaling       | PASS                                                    | local Alice/Bob detector-dephasing channel does not alter remote reduced state or remote outcome marginal                                        | 3.290e-16 |
| singlet_chsh_and_guardrails               | PASS                                                    | singlet reaches Tsirelson while product/classical/local guardrails do not fake Bell violation                                                    | 4.441e-16 |
| claim_level                               | BOUNDARY_MEASUREMENT_CHANNELS_PASS_DYNAMICS_NOT_DERIVED | boundary singlet and local measurement-channel grammar are internally consistent and no-signaling; physical creation/update dynamics remain open |           |

## V11.2M - Naive Born relaxation attempt

| Gate                                             | Status         | Meaning                                                                  | Value |
|--------------------------------------------------|----------------|--------------------------------------------------------------------------|-------|
| integrable_nonborn_guardrail_no_magic_relaxation | PASS_GUARDRAIL | stationary/integrable non-Born ensemble does not magically relax to Born | 1     |

| Gate                                        | Status                                            | Meaning                                                                                                 | Value    |
|---------------------------------------------|---------------------------------------------------|---------------------------------------------------------------------------------------------------------|----------|
| born_equivariance_control_in_mixing_field   | CHECK                                             | Born-initialized ensemble remains close to Born density in multimode dent flow                          | 0.180934 |
| coarse_grained_relaxation_in_multimode_flow | DIAGNOSTIC_OPEN                                   | non-Born ensemble shows coarse-grained H decrease and correlation improvement only in mixing dent field | 9.89862  |
| strong_nonborn_relaxation_stress            | DIAGNOSTIC_OPEN                                   | stronger non-Born bias stress test; relaxation may be partial or absent                                 | 3.07947  |
| born_origin_claim_level                     | BORN_RELAXATION_MECHANIS<br>M_SEED_NOT_DERIVATION | coarse-grained relaxation can be tested as a mixing mechanism; Born origin is not closed by this toy    | 6.48905  |

## V11.2M2 - Equivariance-safe Born relaxation retest

| Gate                                             | Status                                                          | Meaning                                                                                                       | Value    |
|--------------------------------------------------|-----------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------|----------|
| integrable_nonborn_guardrail_no_magic_relaxation | PASS_GUARDRAIL                                                  | periodic stationary/integrable non-Born ensemble does not magically relax to Born                             | 0.995991 |
| born_equivariance_preflight_multimode            | PASS                                                            | Born-initialized ensemble remains close to $ \Psi _2$ under adaptive unclipped multimode torus flow           | 0.980289 |
| adaptive_node_handling_diagnostics               | PASS_DIAGNOSTIC                                                 | adaptive substepping replaced velocity clipping; node-risk/speed diagnostics recorded                         | 4        |
| coarse_grained_relaxation_if_preflight_passes    | DIAGNOSTIC_OPEN                                                 | non-Born H decrease is interpreted only if Born equivariance preflight passes                                 | 0.894258 |
| strong_nonborn_relaxation_if_preflight_passes    | DIAGNOSTIC_OPEN                                                 | strong non-Born stress interpreted only if Born equivariance preflight passes                                 | 0.908832 |
| claim_level                                      | BORN_PREFLIGHT_PASS_RELAX<br>ATION_DIAGNOSTIC_INTERPRET<br>ABLE | M2 separates numerical equivariance preflight from Born-origin relaxation; no Born-rule derivation is claimed |          |

## V11.2M3 - Mixing-linked Born relaxation scan

| Gate                                             | Status                                           | Meaning                                                                                      | Value    |
|--------------------------------------------------|--------------------------------------------------|----------------------------------------------------------------------------------------------|----------|
| stationary_nonborn_guardrail_no_magic_relaxation | PASS_GUARDRAIL                                   | mode_count=1 non-Born distribution should not magically relax under simple torus flow        | 0.995429 |
| born_preflight_pass_fraction                     | PASS                                             | fraction of mode scans whose Born-control passes before interpreting relaxation              | 0.833333 |
| mixing_relaxation_trend                          | PASS_MECHANISM                                   | among preflight-passed modes, $H_f/H_0$ should decrease as mixing proxy increases            | -1       |
| best_relaxation_ratio                            | PASS_MECHANISM                                   | best preflight-valid non-Born $H_f/H_0$ ratio in scan                                        | 0.390791 |
| claim_level                                      | MIXING_LINKED_BORN_RELAXA<br>TION_MECHANISM_SEED | M3 tests whether Born relaxation strengthens with mixing; no Born-rule derivation is claimed |          |

## V11.2N - Native lag-current bridge cascade

| Gate                                           | Status                                   | Meaning                                                                                                                                                       | Value   |
|------------------------------------------------|------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|
| density_only_lag_does_not_fake_quantum_current | PASS_GUARDRAIL                           | density residual -grad $\Lambda_{\rho}$ should not reproduce the phase/current guidance field                                                                 | 0       |
| action_phase_lag_reconstructs_quantum_current  | CHECK                                    | maintained action/clock phase lag gives -grad(-S)=grad S= $v_Q$ at representation level                                                                       | 0.18597 |
| settlement_residual_direct_velocity            | DIAGNOSTIC_OPEN                          | finite lag residual of action phase is diagnostic; direct equality to $v_Q$ is not expected and remains V12 theory work                                       | 0       |
| claim_level                                    | NATIVE_LAG_CURRENT_BRIDGE<br>_NOT_CLOSED | V11.2 current bridge must run through maintained action phase/clock holonomy; density dents alone are insufficient; V11.1B settlement derivation remains open |         |

## V11.2N2 - Action-phase / quantum-current bridge audit

| Gate                                    | Status | Meaning                                                                                     | Value     |
|-----------------------------------------|--------|---------------------------------------------------------------------------------------------|-----------|
| plane_wave_phase_current_identity       | PASS   | phase-gradient extraction reproduces $v_Q$ for exact plane wave                             | 2.638e-13 |
| no_node_two_mode_phase_current_identity | PASS   | phase-gradient extraction reproduces $v_Q$ in controlled no-node interference state         | 1.755e-06 |
| weak_node_multimode_masked_bridge       | PASS   | action-phase bridge works in weak multimode case after excluding invalid node neighborhoods | 4.302e-06 |
| strong_node_multimode_masked_bridge     | PASS   | gnarly multimode case is interpretable only on node-masked phase-valid domains              | 1.618e-05 |

| Gate        | Status                                                           | Meaning                                                                                          | Value |
|-------------|------------------------------------------------------------------|--------------------------------------------------------------------------------------------------|-------|
| claim_level | ACTION_PHASE_CURRENT_BRIDGE_VALIDATED_CONTROLLED_AND_NODE_MASKED | N2 audits action-phase current extraction only; V11.1B settlement-to-action dynamics remain open |       |



# Reupload / continuation instructions

| Action                      | Instruction                                                                                                                                       |
|-----------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|
| If reuploading later        | Use this PDF as the stable narrative ledger. Also upload any target verdict CSVs or repro scripts if the next task requires exact recomputation.  |
| If improving V11.2 document | Start from the master ledger table. Promote only closed representation passes; keep mechanism seeds and open gates in their own sections.         |
| If building repro pack      | Include scripts v11_2a through v11_2n2 plus each output directory/verdict CSV. Mark M and N as superseded but historically important.             |
| If writing claims           | Use the claim_level strings exactly. Do not turn MECHANISM_SEED into DERIVATION. Do not turn PASS_PURE_STATE_ONLY into full measurement dynamics. |

## Recommended next strategic move

The V11.2 quantum-foundations sector is now coherent enough to park temporarily. The highest-leverage next work is empirical or consolidation: update the V11.2 sector document, bundle the V11.2 repro scripts, then return to the TNG/DESI crossing pipeline and/or SPARC journal packaging. If continuing V11.2, the deepest open bridge is V11.1B settlement dynamics -> maintained action phase S.